

Netflix Content Analytics

GitHub Project Documentation

Business Intelligence & Data Science — Exploratory Data Analysis

Dataset: 8,807 Netflix title records

Project focus: Catalog composition, geography, ratings, genres, TV-show seasons and data quality

Prepared as a repository-ready technical document based on the supplied project presentation and Jupyter Notebook.

1. Project Overview

This project analyzes Netflix title data to understand catalog composition, geographic presence, ratings, genres, TV-show season patterns, release-year trends, and data-quality considerations. The supplied presentation describes the work as a practical introductory Business Intelligence and Data Science EDA workflow.

2. Business Objectives

- Analyze the Movie vs. TV Show mix.
- Identify countries leading in content volume.
- Understand dominant ratings and genres.
- Explore TV-show season distribution.
- Examine release-year patterns.
- Identify data-quality issues that matter for production analytics.
- Translate EDA findings into a BI dashboard and future ML phase.

3. Dataset Summary

Metric	Value	Interpretation
Total titles	8,807	Records analyzed
Movies	6,131 (69.6%)	Movie-dominated catalog
TV Shows	2,666 (30.3%)	Smaller but significant share
India titles	972	Second-largest country group
Top country	United States — 3,642	Dominant content origin
Peak release year	2018 — 1,146 titles	Highest listed output

4. Repository Structure

```
netflix-content-analytics/  
├── data/  
│   └── netflix_titles.csv  
├── notebooks/  
│   └── Netflix_data(1).ipynb  
├── reports/  
│   └── Netflix_PPT.pdf  
├── README.md  
├── requirements.txt  
└── LICENSE
```

The tree above is a recommended GitHub organization based on the supplied notebook and presentation. The source files supplied for this documentation are the Jupyter Notebook and project presentation.

5. Tools & Technologies

Detected from the supplied notebook: pandas, numpy, matplotlib, seaborn. The presentation describes the workflow as Load → Clean → Transform → Visualize.

Notebook imports:

```
import pandas as pd  
import numpy as np  
import seaborn as sns
```

```
import matplotlib.pyplot as plt
import warnings
```

6. EDA Workflow

Stage	Project activity
Load	Load the Netflix title dataset into the analysis environment.
Clean	Handle missing values and inspect data-quality issues.
Transform	Convert date_added to datetime and attempt Year/Month feature extraction.
Explore	Use value counts and descriptive summaries to understand distributions.
Visualize	Use bar charts, line charts, pie charts and a missing-value heatmap.
Interpret	Translate patterns into business KPIs, dashboard requirements and future ML opportunities.

7. Key EDA Findings

Catalog composition: Movies account for nearly 70% of records: 6,131 movies versus 2,666 TV shows.

Geographic distribution: The United States leads with 3,642 records, followed by India (972) and the United Kingdom (418). Japan (244) and South Korea (199) are notable Asian markets.

Ratings: TV-MA (3,205) and TV-14 (2,157) are the most frequent ratings, indicating a strong mature-audience presence.

Genres: International Movies, Dramas, Comedies, International TV Shows and Documentaries are highlighted as the leading listed genres.

TV seasons: Single-season shows dominate with 1,793 titles, followed by 421 with two seasons and 356 with three or more seasons.

Release years: The dataset spans 1925–2021, with 2018 as the peak listed release year at 1,146 titles and a visible decline from 2020 onward.

8. Data Quality & Limitations

Field	Missing values / issue	Implication
Director	2,634 missing	Mode imputation can distort frequency-based director analysis.
Cast	825 missing	Actor/cast analysis may be incomplete.
Country	831 missing	Geographic analysis has incomplete records.
date_added	10 missing	Datetime-derived features require validation.
Rating	4 missing	Small but non-zero gap in audience segmentation.
Duration	3 missing	Small gap for duration analysis.

The project presentation specifically warns that director ranking should not be treated as a true ranking after mode imputation. This is an important limitation to document in the GitHub README.

9. Suggested GitHub README

```
# Netflix Content Analytics

## Overview
Exploratory Data Analysis of 8,807 Netflix title records covering catalog composition,
country distribution, ratings, genres, TV-show seasons and release-year patterns.

## Objectives
- Compare Movies and TV Shows
```

```
- Analyze geographic content distribution
- Identify dominant ratings and genres
- Explore TV-show season patterns
- Assess data quality
- Define dashboard and future ML opportunities

## Workflow
Load → Clean → Transform → Visualize → Interpret

## Key Results
- 6,131 Movies and 2,666 TV Shows
- United States: 3,642 records
- India: 972 records
- TV-MA: 3,205 titles
- TV-14: 2,157 titles
- 2018: peak release year with 1,146 titles
- Single-season TV shows: 1,793

## Future Work
Build a BI dashboard, automate the ETL workflow, and extend the project
with recommendation, clustering, or demand-forecasting models.
```

10. Deployment Roadmap

- Version-control the notebook, README, requirements, data dictionary and presentation.
- Create a repeatable ETL pipeline: validation → cleaning → feature engineering.
- Add automated data-quality checks for missingness and datetime parsing.
- Publish a BI dashboard with filters for country, genre, rating, content type, release year and duration.
- Schedule dashboard refresh after successful ETL validation.
- Treat recommendation engines, clustering and demand forecasting as a separate ML phase.

11. Conclusion

The Netflix Content Analytics project demonstrates an introductory end-to-end EDA workflow with clear business dimensions. The analysis identifies a movie-dominated catalog, strong representation from the United States and India, concentration in mature ratings, and a strong single-season TV-show pattern. The next practical step is to operationalize these findings in a BI dashboard while strengthening data validation and missing-value handling before production deployment.

12. Source Materials

This documentation was created from the supplied **Netflix PPT.pdf** and **Netflix data(1).ipynb**. Project facts and terminology in this document are grounded in those materials.